Software Engineering

LAB -1

Name: Avrit Sharma
SRN: PES2UG24CS100

PROBLEM STATEMENT #34: DIGITAL GIFT CARD & LOYALTY POINT EXCHANGE

Requirements Table

Actors

• Account Holder: aggregates points, converts them to exchange credits, views wallet information, and redeems digital gift cards.

• Merchant Partner: supplies loyalty-point information that is validated before conversion.

Done by -
Avrit Sharma
PES2UG24CS100

Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall validate merchant loyalty points, convert them into universal exchange tokens, and generate single-use digital voucher codes.	High	Pass: Voucher code is generated with a cryptographic checksum after valid point conversion. Fail: Expired or duplicate voucher redemption is accepted.	Provides the core conversion and voucher-generation workflow while enforcing the stated anti-fraud requirement.
FR-002	Functional	The system shall allow an Account Holder to aggregate eligible brand loyalty points from participating Merchant Partners into the digital wallet.	High	Pass: Points from a participating merchant are added to the Account Holder's wallet balance after successful validation. Fail: Invalid points are added to the wallet.	Enables the central wallet function of consolidating loyalty points from multiple brands.
FR-003	Functional	The system shall allow an Account Holder to redeem universal exchange credits for a selected digital gift card and lock the voucher before completing redemption.	High	Pass: The selected voucher is locked before redemption is completed. Fail: A voucher remains redeemable by another transaction after it has been locked.	Supports digital gift-card redemption and prevents concurrent or repeated use of a voucher.
FR-004	Functional	The system shall allow a Merchant Partner to provide loyalty-point information for validation before the points are converted into universal exchange credits.	Medium	Pass: Only point information that passes validation proceeds to conversion. Fail: Unverified merchant points are converted into exchange credits.	Ensures that exchange credits are created only from valid merchant loyalty points.
FR-005	Functional	The system shall allow an Account Holder to view the current universal exchange-credit balance and the status of completed point-conversion and gift-card-redemption transactions.	Medium	Pass: The wallet displays the current balance and corresponding transaction status after a completed operation. Fail: The displayed balance or status does not reflect the completed transaction.	Gives users visibility into wallet value and the outcome of their exchanges and redemptions.

Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	Point conversion and gift card redemption operations must process within 250 ms with zero ledger balance drift.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Either operation exceeds 250 ms or produces ledger balance drift.	Ensures responsive transactions while maintaining financial consistency and integrity.
NFR-002	Security	The system shall enforce single-use voucher locking so that an expired, locked, or previously redeemed digital gift-card voucher cannot be redeemed again.	High	Pass: A second or duplicate redemption attempt is rejected and the voucher remains locked. Fail: The same voucher is successfully redeemed more than once.	Provides the anti-fraud protection explicitly required for digital gift-card redemption.

Done by -
Avrit Sharma
PES2UG24CS100

Use-Case Flow Specification

Problem Statement #34: Digital Gift Card & Loyalty Point Exchange

Use case name: Redeem Digital Gift Card

Primary Actor: Account Holder

Preconditions

The Account Holder is authenticated and has sufficient universal exchange credits for the selected gift card.

The selected digital gift card is available for redemption.

The voucher associated with the redemption is not expired, locked, or previously redeemed.

The wallet balance is current and available for the transaction.

Postconditions

The required exchange credits are deducted from the Account Holder's wallet.

The digital gift-card voucher is locked and marked as redeemed for the completed transaction.

A redemption transaction record is stored with its status.

Main Success Scenario

1. Account Holder selects a digital gift card and chooses the Redeem option.

2. System displays the gift-card details, required exchange credits, and redemption confirmation.

3. Account Holder confirms the redemption.

4. System verifies that the Account Holder has sufficient universal exchange credits.

5. System validates that the selected voucher is eligible for redemption.

6. System locks the voucher to prevent duplicate or concurrent redemption.

7. System deducts the required exchange credits from the Account Holder's wallet.

8. System marks the voucher as redeemed and records the transaction.

9. System displays the digital gift-card voucher code and redemption confirmation to the Account Holder.

10. Use case ends successfully.

Done by -
Avrit Sharma
PES2UG24CS100

Alternate Flow

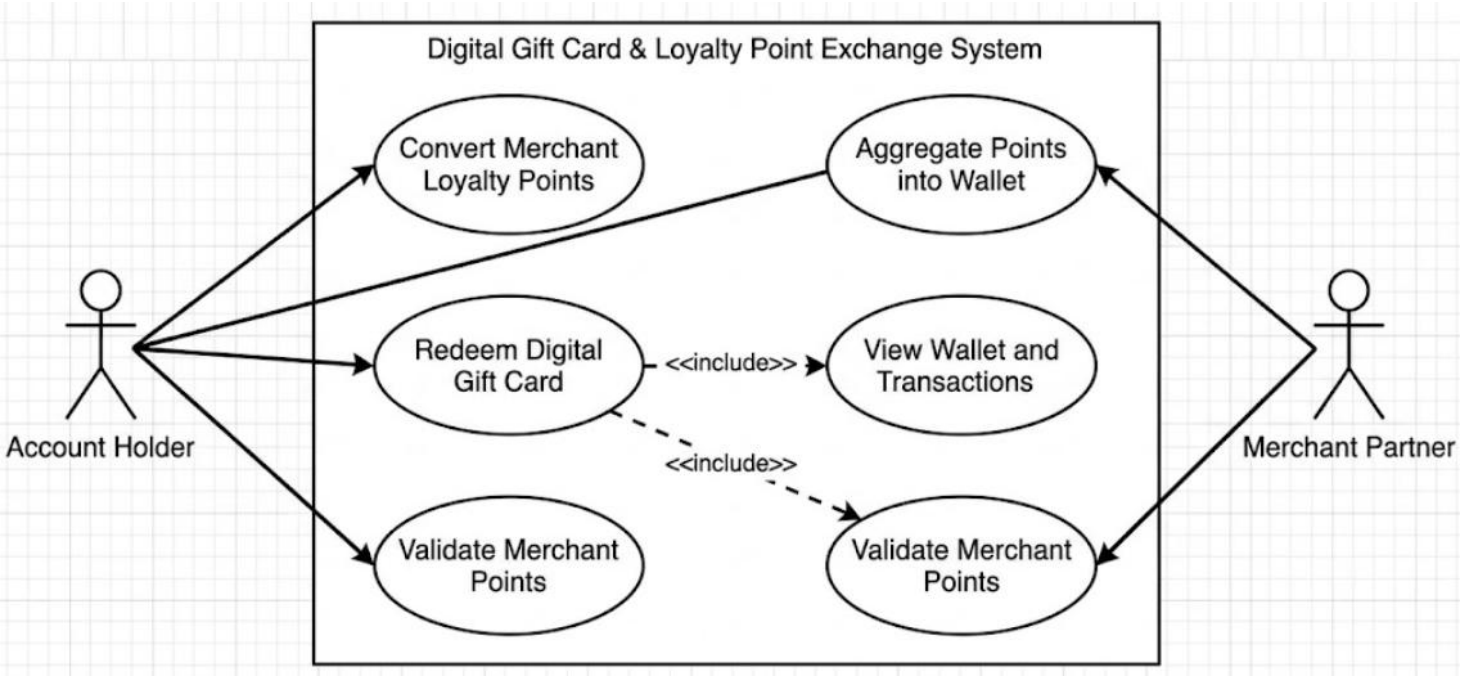
5a. Voucher Invalid, Expired, or Already Redeemed

5a1. System detects that the selected voucher is expired, locked, or previously redeemed.

5a2. System rejects the redemption and does not deduct exchange credits from the Account Holder's wallet.

5a3. System displays a voucher-unavailable message and prompts the Account Holder to select another gift card.

UML Use-Case Diagram



Done by -
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